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Roll No .....

**EX-7203-GS**

**B.E. VII Semester**

Examination, December 2020

**Grading System (GS)**

**Computer Networks**

**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Draw the reference architecture of ISO-OSI model? Also, explain the function of each layer briefly.  
b) State the various applications of computer network?
2. a) What is circuit switched network? How communication is established in these networks?  
b) Explain with examples different types of guided and unguided medias in computer network?
3. What is bit stuffing. If the start and end header is 100001 and the following data stream is to be bit stuffed - 100110001100001110000011. What will be the frame after bit stuffing?
4. a) Measurement of slotted aloha channel with an infinite number of user show that 10 percent of slots are idle.  
i) What is the channel load G?  
ii) What is the through put?  
iii) Is the channel under loaded and over loaded.  
b) Compare the performance and efficiency of Go-back-N and selective repeat protocols?

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5.
  - a) Discuss various design issues of network layer?
  - b) Explain various routing techniques?
6.
  - a) Distinguish between :
    - i) Static and dynamic routing
    - ii) Centralized and distributed routing
  - b) What are gateway design issues X25 internet working?
7.
  - a) Differentiate between ISDN and B-ISDN?
  - b) Explain the working of ATM adaptation layer and transport layer?
8. Write short notes on (Any two)
  - a) Session layer design issues
  - b) Deadlock in Ethernet
  - c) Centralized isolated distributed hierarchical broadcast.

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